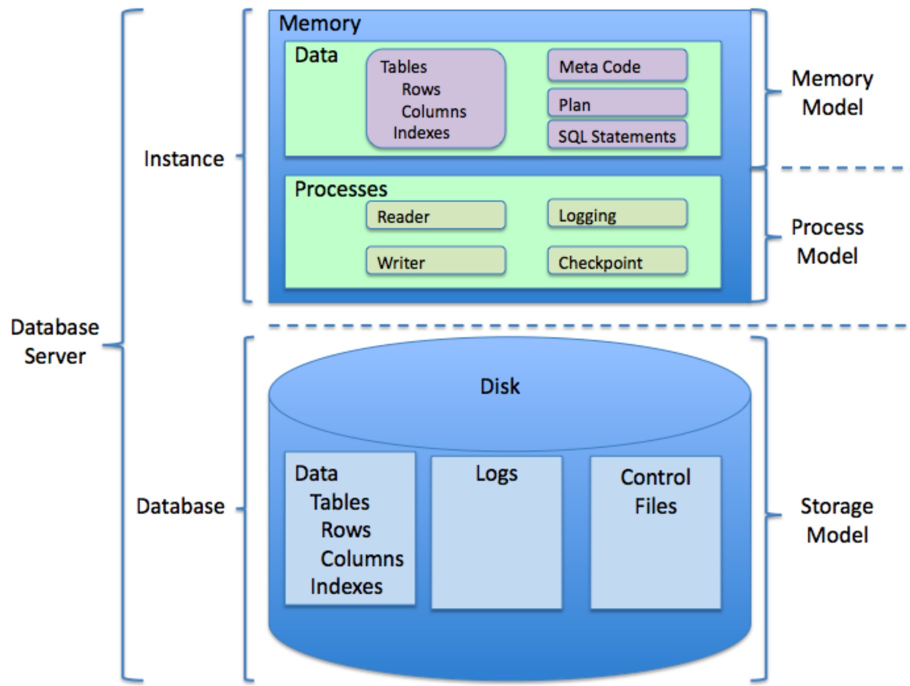
Database Management System



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Database Design

- Focuses on the design of the database structure that will be used to store and manage end-user data
- Well-designed database
  – Facilitates data management
  – Generates accurate and valuable information
- Poorly designed database causes difficult-to-trace errors

Evolution of File System Data Processing

Manual File Systems

Accomplished through a system of file folders and filing cabinets

Computerized File Systems

Data processing (DP) specialist: Created a computer-based system that would track data and produce required reports

File System Redux: Modern End-User Productivity Tools

Includes spreadsheet programs such as Microsoft Excel

Basic File Terminology

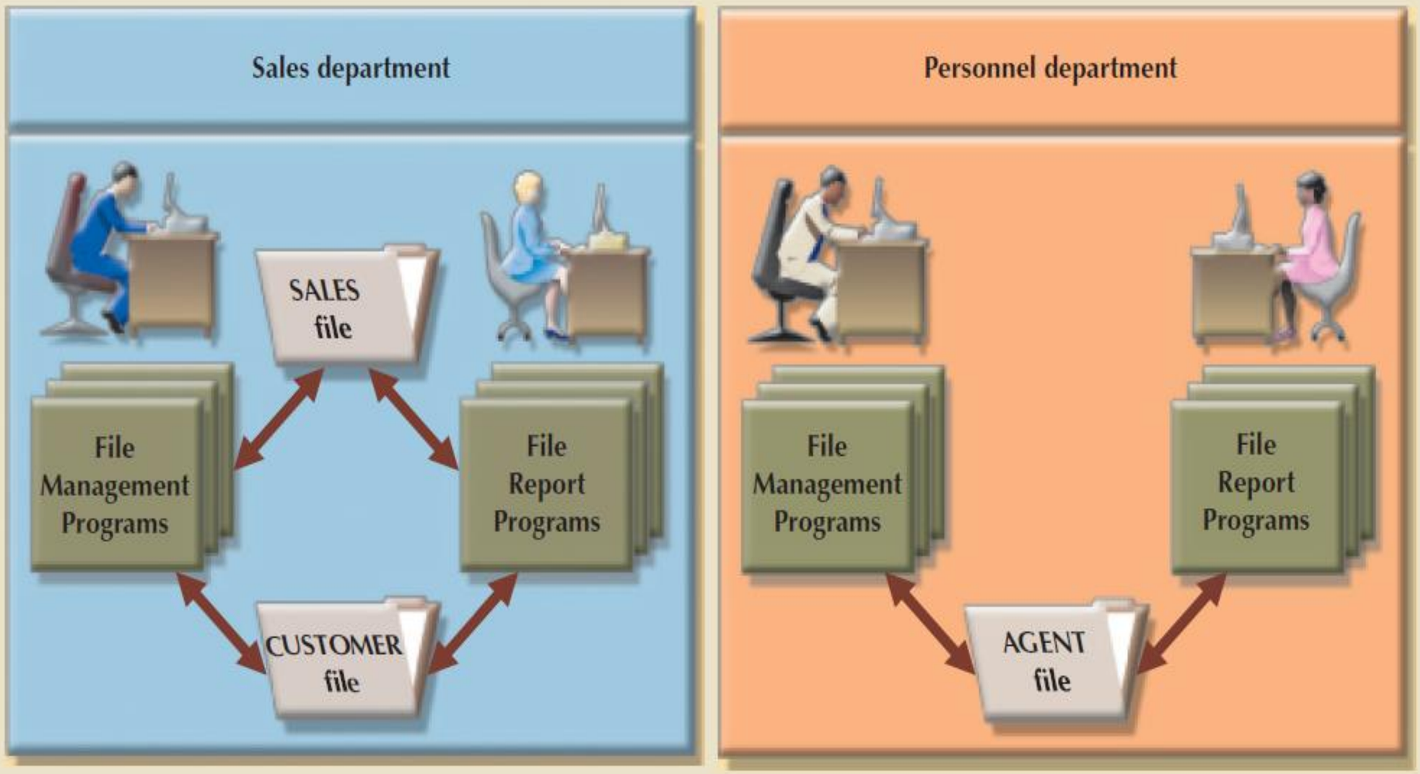
TABLE 1.2

BASIC FILE TERMINOLOGY

TERM	DEFINITION
Data	Raw facts, such as a telephone number, a birth date, a customer name, and a year-to-date (YTD) sales value. Data have little meaning unless they have been organized in some logical manner.
Field	A character or group of characters (alphabetic or numeric) that has a specific meaning. A field is used to define and store data.
Record	A logically connected set of one or more fields that describes a person, place, or thing. For exam-ple, the fields that constitute a record for a customer might consist of the customer's name, address, phone number, date of birth, credit limit, and unpaid balance.
File	A collection of related records. For example, a file might contain data about the students currently enrolled at Gigantic University.

A Simple File System

FIGURE 1.8 A SIMPLE FILE SYSTEM



Problems with File System Data Processing

- Lengthy development times
- Difficulty of getting quick answers
- Complex system administration
- Lack of security and limited data sharing
- Extensive programming

Structural Dependence

- **Structural dependence**: Access to a file is dependent on its own structure
  - All file system programs are modified to conform to a new file structure
- **Structural independence**: File structure is changed without affecting the application's ability to access the data

Structural Dependence Implications

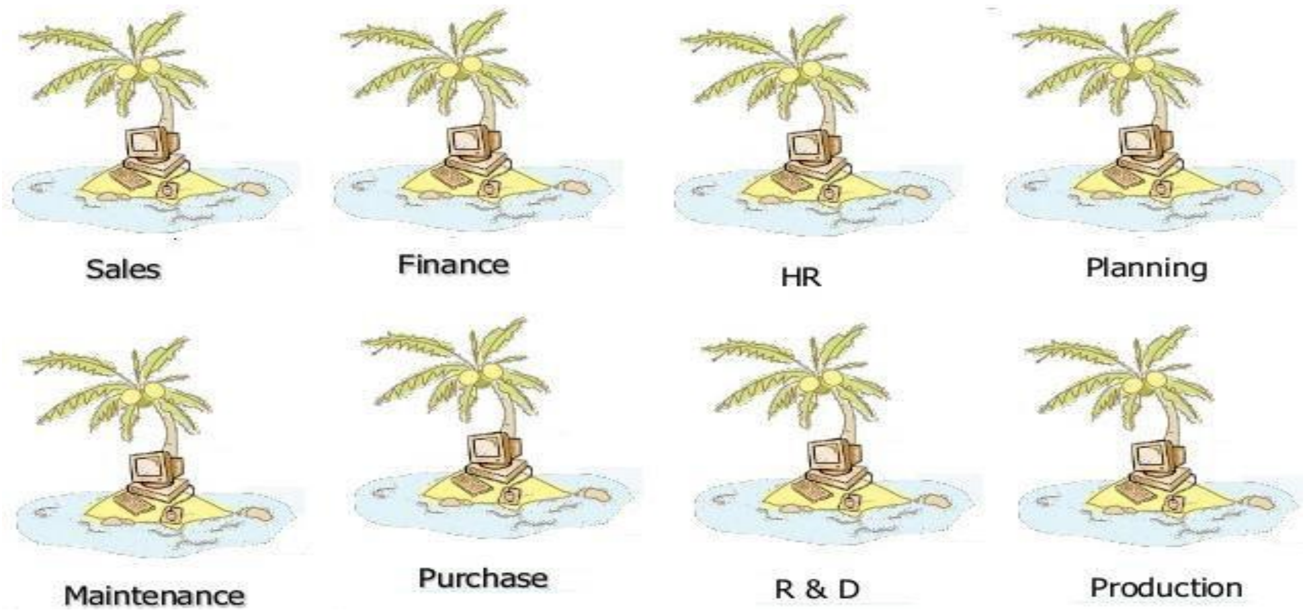
- Adding a customer date of birth field to the customer file would require a program that
  - Reads the original file
  - Transforms the original data to conform to the new structure's storage requirement
  - Writes the transformed data into the new file structure
- Because the file system application programs are affected by changes in the file structure, they exhibit structural dependence

Data Dependence

- Data dependence
  - Data access changes when data storage characteristics change
- Data independence
  - Data storage characteristics is changed without affecting the program's ability to access the data

Data Redundancy

- Unnecessary storage of same data at different places which may result in having different versions of the same data



Data Redundancy Implications

- Poor data security
- Data inconsistency
- Increased likelihood of data-entry errors when complex entries are made in different files
- **Data anomaly**: Develops when not all of the required changes in the redundant data are made successfully

Types of Data Anomaly

- Update Anomalies
- Insertion Anomalies
- Deletion Anomalies

Advantages of Using DBMS Approach

- Improved Data Sharing: DBMS provides coherent view of data to End-Users of different departments with respect to their requirements.
- Improved Data Security: DBMS provides a framework for better enforcement of data privacy and security policies.
- Better Data Integration: DBMS provides an integrated view of organizations operations and clearer view of the big picture. It becomes much easier to see how actions in one segment of the company affect other segments.

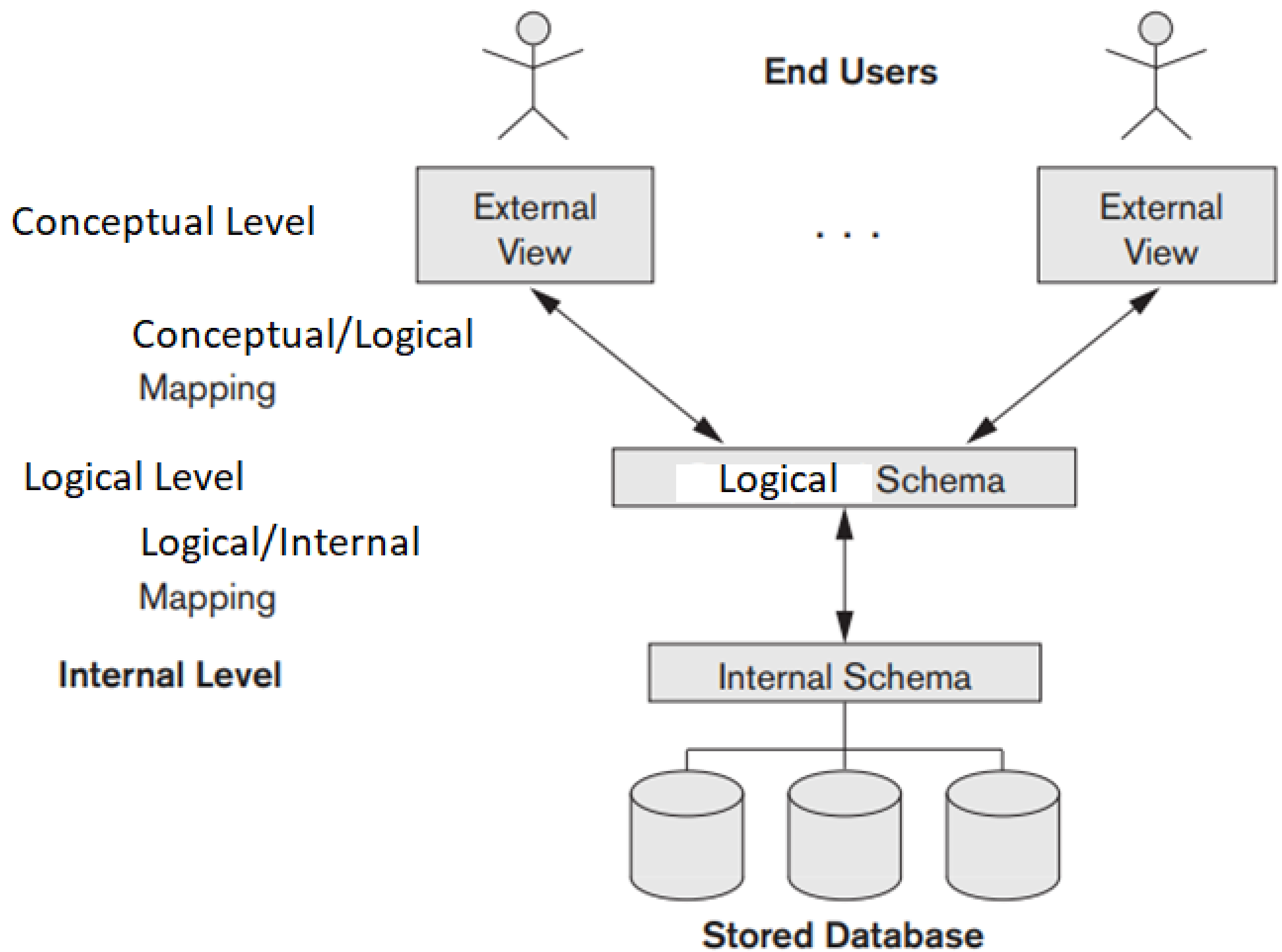
Advantages of Using DBMS Approach

- Minimized Data Inconsistency: Data inconsistency exists when different versions of the same data appear in different places. DBMS keeps only one version of data (at one place logically) with most latest updates. One logical data-file removes the problem of inconsistency.
- Improved Data Presentation w.r.t user requirements which increases end-user productivity and quality of decisions.

Disadvantages of using DBMS

- Increased costs
  - The cost of maintaining specialized hardware, software and human resource can be substantial
- Vendor dependence
  - Given the heavy investment in Database system companies might be reluctant to change DBMS vendors. As a result customers are limited in their choice of database system components
- Frequent upgrade/replacement cycles
  - DBMS vendors upgrade their products by adding new functionality. Such new features often require software / hardware upgrades. Upgrades cost money in purchasing and user training.

The 3-Schema Architecture



The 3-Schema Architecture

- The goal of the 3-schema architecture is to separate the user applications from the physical database. In this architecture schemas can be defined at the following 3 levels
  - Internal level has internal schema: describes the physical storage structure of the database. The internal schema uses a physical data model and describes the complete details of data storage and access paths for the database.

The 3-Schema Architecture

– Logical level has a logical schema, which describes the structure of the whole database for a community of users. It hides the details of physical storage structures and concentrates on describing entities, data types, relationships, user operations, and constraints. A representational data model is used to describe the logical schema when a database system is implemented.

The 3-Schema Architecture

- External or User Views or conceptual schema: it describes the part of the database that a particular user group is interested in and hides the rest of the database from that user group. It is implemented using a representational data model.